

ABCD-LINK: Annotation Bootstrapping for Cross-Document Fine-Grained Links

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Abstract

Understanding fine-grained relations between documents is crucial for many application domains. However, the study of automated assistance is limited by the lack of efficient methods to create training and evaluation datasets of cross-document links. To address this, we introduce a new domain-agnostic framework for selecting a best-performing approach and annotating cross-document links in a new domain from scratch. We first generate and validate semi-synthetic datasets of interconnected documents. This data is used to perform automatic evaluation, producing a shortlist of best-performing linking approaches. These approaches are then used in an extensive human evaluation study, yielding performance estimates on natural text pairs. We apply our framework in two distinct domains – peer review and news – and show that combining retrieval models with LLMs achieves 78% link approval from human raters, more than doubling the precision of strong retrievers alone. Our framework enables systematic study of cross-document understanding across application scenarios, and the resulting novel datasets lay foundation for numerous cross-document tasks like media framing and peer review. We make the code, data, and annotation protocols openly available.¹

1 Introduction

Documents rarely exist in isolation. In many practical scenarios, understanding one document requires reasoning over its relationships with others. Fact-checkers might trace a claim to the specific sentences across multiple articles that support or contradict it (Thorne et al., 2018; Wadden et al., 2020; Chen et al., 2024b). Authors and meta-reviewers reading peer reviews need to connect reviewer comments to the paper to decide on the course of action (Kuznetsov et al., 2022; D’Arcy et al., 2024). News

¹<https://anonymous.4open.science/r/ABCD-LINK-C999>

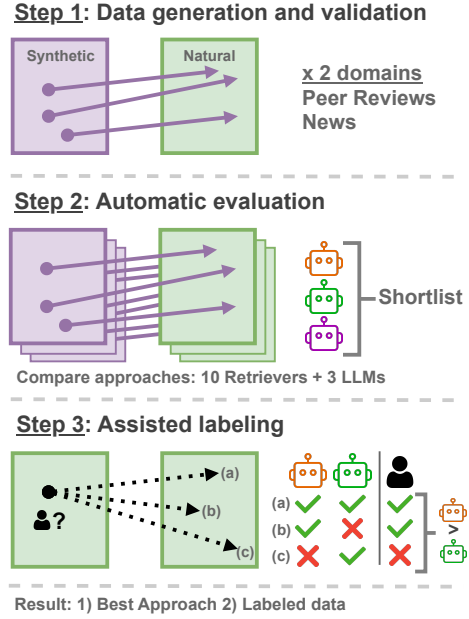


Figure 1: Framework overview.

analysts might use paraphrased or ideologically re-framed sentences across different outlets to gain insights into reporting bias and source alignment (Giorgi et al., 2023). Yet finding such fine-grained relations can be cognitively demanding, motivating the need for machine assistance. Following Kuznetsov et al. (2022), we call fine-grained relations between documents *links*, and the general task of identifying them *linking*. While automatic linking holds great potential, the progress has been limited due to the manual effort required to annotate and evaluate links, and the diversity of link types across domains. Because of this, labeled corpora of links are scarce, preventing the development and evaluation of automated linking approaches at scale.

To move beyond the bottleneck of full manual annotation, we propose a general framework for bootstrapping sentence-level cross-document linking without the need for pre-existing labeled data

(Figure 1). While links might be hard for humans to *detect*, linked documents are easier to *generate* with state of the art LLMs. Building upon this insight, we (1) propose a method to generate and validate semi-synthetic corpora of linked documents. We use this data to (2) automatically evaluate a wide range of approaches to arrive at a shortlist of best-performing linking approaches. Finally, we use the best-performing approaches to (3) assist humans in labeling natural, non-synthetic document pairs in a large-scale human-in-the-loop evaluation study. As a result, our framework allows practitioners to efficiently select the best linking approach given a domain and link type, while producing manually labeled linking data for downstream applications.

We assess our framework by applying it in two distinct domains, peer reviews and news articles, each with its own task-specific link type and underlying data. In total, we contribute:

- A novel framework for efficient human-in-the-loop evaluation and labeling of links;
- A data generation approach for creating semi-synthetic datasets of linked documents in the domain of interest, incl. human validation;
- A large-scale analysis of state-of-the-art linking approaches, incl. a novel human evaluation protocol;
- Two new manually annotated non-synthetic corpora of cross-document links in peer review and news.

We release the code, datasets, and annotation protocols to support future research on sentence-level cross-document linking and its applications.

2 Related Work

Numerous NLP tasks study or rely on **cross-document relations**, including cross-document coreference (Miyabe et al., 2008; Cybulska and Vossen, 2014; Ravenscroft et al., 2021), fact-checking (Thorne et al., 2018; Wadden et al., 2020; Chen et al., 2024b), scholarly peer review analysis (Kuznetsov et al., 2022; D’Arcy et al., 2024), and citation/source detection (Syed et al., 2023; Liang et al., 2024). While prior work focuses on limited relation types and relies on manual labeling, we contribute a general, structured framework for bootstrapping linking from scratch in new domains and for new link types.

From a **machine-assisted annotation** perspective, prior work reduces labeling effort by pre-selecting candidate links with heuristics (Raven-

croft et al., 2021), or simple retrieval methods, e.g. cosine similarity plus a fixed cutoff (Kuznetsov et al., 2022). Our framework generalizes this idea: we automatically benchmark various zero-shot approaches, and validate the results through human evaluation to select the best approach for the task and domain at hand.

The use of **synthetic data** in NLP is increasingly common. Recent work has demonstrated the promise of LLMs for data augmentation across classification and reasoning tasks (Ding et al., 2024). Liu et al. (2022) show that machine-human collaboration can produce diverse and challenging natural language inference examples, and Zhao et al. (2025) use LLMs to decontextualize event mentions and create denser cross-document event coreference datasets. Furthermore, Møller et al. (2024) and Kazemi et al. (2025) find that synthetic data can match or complement human-labeled corpora in low-resource settings. To the best of our knowledge, we are the first to use synthetic data to study cross-document links. The use of synthetic data carries risks (Van Breugel et al., 2023; Lu et al., 2023). To mitigate them, our framework only uses synthetic data to automatically evaluate different linking approaches, employs a human validation protocol, and supplements evaluations on synthetic data with focused human evaluation, contributing to best practices in the use of synthetic data in NLP.

Human judgment is essential for evaluating NLP systems, yet difficult to elicit reliably. There is growing interest in structured **human evaluation protocols**, particularly for tasks involving model-assisted annotation or subjective decision-making. Prior work examined inter-annotator calibration over time (Uma et al., 2021), preference-based assessment of model outputs (Clark et al., 2021), and the influence of interface design on annotation quality (Klie et al., 2018). In the context of LLM evaluation, human-in-the-loop setups have proven essential for capturing fine-grained distinctions and improving label quality (Zhou et al., 2023). To human evaluation our framework contributes a novel and efficient annotation protocol that combines candidate pre-selection with a ranking-based comparison for a reliable assessment of top-performing approaches.

3 Setup

To trade off between task complexity and utility, we focus on *sentence-level linking*, cast as a bi-

nary classification task. Given two documents, the source document $A = \{a_1, \dots, a_N\}$ and the target document $B = \{b_1, \dots, b_M\}$, the goal is to predict for each cross-document sentence pair (a_i, b_j) whether a relation r holds between them. The definition accommodates both directed links (e.g. a review sentence criticizing a claim in a paper) and undirected links (e.g. sentences in two new articles addressing the same fact). Linking can be seen as a special case of information retrieval, where a_i serves as a query to retrieve relevant sentences from B . However, linking is constrained by a task-specific relationship holding between sentences, and might require documents A and B for contextualizing the linking decisions.

There are many potential approaches to perform automatic linking in a zero-shot fashion, from simple relation-agnostic cosine similarity to classification with state-of-the-art LLMs. Yet, labeled datasets of links are scarce, and *evaluating and comparing* the performance of different linking approaches on a wide range of link types and domains via human evaluation is infeasible. Our framework addresses this limitation by generating and validating semi-synthetic data (Section 4) to arrive at a shortlist of best-performing linking approaches (Section 5) which are then used in a focused human evaluation and annotation study (Section 6).

We showcase our framework on two distinct practical scenarios: in REVIEWS, we link peer reviews to their papers, in NEWS we link pairs of news articles. While the method is link-type agnostic, we focus on task-specific link types for each domain. In REVIEWS, we define a link as a sentence in a review that comments on, critiques, or supports a specific sentence in the corresponding paper. In NEWS, a link connects two sentences from different articles that convey the same or closely related factual content, typically reflect semantic equivalence or paraphrastic overlap, even if the framing or tone differs between sources.

4 Synthetic data

4.1 Generation

We hypothesize that while links in existing document pairs can be time-consuming to annotate, they are relatively easy to generate. Based on this intuition, we construct a semi-synthetic linking dataset for each of our target application domains using a non-synthetic, natural document as target, and prompting an LLM to generate a

Stat	NEWS	NEWS	REVIEWS	REVIEWS
	ECB+	SYNTH	SYNTH	F1000
Doc Pairs	2505	346	200	211
Number of Links	9383	2323	2181	1205
Avg. Sents (Src)	17.7	12.0	10.5	22.4
Avg. Sents (Tgt)	18.2	13.7	130.8	145.0
Avg. Links (Src)	2.06	4.97	5.14	4.84
Avg. Links (Tgt)	3.75	6.71	10.9	5.71

Table 1: Dataset statistics: number of document pairs, number of links, average number of sentences in source and target documents, average number of linked sentences in source and target document.

synthetic source document that links to the target on the sentence level. We use DeepSeek-R1 (DeepSeek-AI et al., 2025), which in our preliminary experiments demonstrated strong reasoning and instruction-following capabilities, resulting in higher-quality synthetic data. While strictly speaking only the source documents in our data are synthetic, for simplicity, we further refer to the resulting data as our *synthetic data*.

NEWS-SYNTH We sample 346 news articles from the WikinewsSum dataset (Calizzano et al., 2022). To reduce noise in the documents, we clean each article using GPT-4o-mini² to remove web-scraping artifacts. We then segment the text into sentences using spaCy³ (Honribal et al., 2020). Next, we prompt the LLM to produce a related article on the same topic, written with a different tone or editorial perspective, and instructing the model to link sentences in the generated article to sentences in the original article, resulting in cross-document links. This is intended to simulate a use case in media analysis where the same events reported by different outlets are compared to detect bias and framing.

REVIEWS-SYNTH We use the most recent version of the NLPeer dataset (Dycke et al., 2023), which includes EMNLP24 submissions pre-segmented into sentences. We select the 200 shortest papers by sentence count and prompt the model to generate a peer review for each. We focus on short papers to mitigate issues with long-context during inference (Levy et al., 2024). The prompt instructs the LLM to write review sentences that are linked to particular sentences in the paper, simulating naturally

²<https://platform.openai.com/docs/models/gpt-4o-mini>

³model: en_core_web_md

occurring reviewer comments. See Appendix L for prompt templates and example links for both domains.

To contextualize our synthetic datasets and experimental results, we derive two additional datasets from existing corpora of cross-document links. **NEWS-ECB** is derived from the ECB+ corpus (Cybulska and Vossen, 2014) designed for cross-document coreference resolution. We repurpose this dataset for sentence-level linking by aligning sentence pairs across documents that share event mentions in the same cluster. **REVIEWS-F1000** leverages the F1000RD dataset introduced by Kuznetsov et al. (2022), which links peer review sentences to their corresponding paper sentences. Although its annotation design restricts the choices of possible links for each review sentence, it is the most structurally similar resource to our task. Dataset conversion details are provided in Appendix A.

Table 1 provides key statistics on the synthetic and converted datasets. To avoid potential bias (Liu et al., 2024), we explicitly exclude DeepSeek-R1 from downstream experiments, and solely use it for synthetic data generation.

4.2 Validation

At this step of the framework, the aim is to produce high-quality source documents that have sentence-level links to a target document. To assess the usability of the data, we conduct further quantitative and qualitative analysis.

Quantitative comparison. We compare synthetic and natural documents using document-level statistics: lexical diversity, subjectivity⁴, and lexical complexity (Flesch-Kincaid Reading Ease score (Flesch, 1948)). In NEWS-SYNTH, synthetic articles are more lexically diverse, similarly subjective, and slightly more lexically complex than their natural counterparts. In contrast, REVIEWS-SYNTH reviews show lower lexical diversity and higher complexity. We attribute this to LLMs’ tendency to use more formal or technical language within the prompt’s length constraints, whereas natural reviews are longer and can use simpler phrasing for communication. Subjectivity remains comparable across domains. Overall, synthetic documents align well with natural ones in most stylistic dimensions. See Appendix B for further details.

⁴<https://huggingface.co/cffl/bert-base-styleclassification-subjective-neutral>

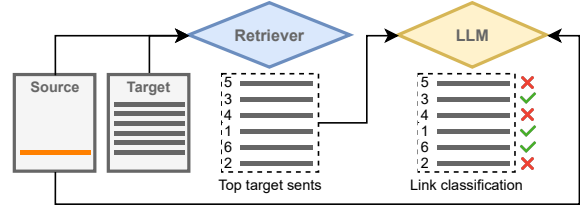


Figure 2: R+LLM Setup.

Human validation To further assess the quality of the synthetic data, we conduct a human validation study for both synthetic datasets. For NEWS-SYNTH, we randomly selected 30 natural news articles and their corresponding synthetic counterparts, yielding 30 document pairs. Four annotators rated each article on a five-point Likert scale for fluency, coherence, realism (i.e., could the article be written by a human), and specificity (i.e., does the article contain specific facts). Synthetic articles match fluency/coherence but score lower on realism and specificity, likely due to LLMs producing more general statements. While these limitations may affect journalistic authenticity, they do not hinder our linking task where coherence is more important, as coherent articles provide better context for reasoning over.

For REVIEWS-SYNTH, we randomly selected 30 synthetic reviews and retrieved natural reviews for the same papers from the NLPeer dataset, again forming 30 document pairs. Three NLP PhD students rated each review on fluency, coherence, helpfulness (i.e., would this review be helpful for the paper’s authors), and specificity (i.e., does the review address concrete parts of the paper). Synthetic reviews were rated higher in fluency and coherence, and lower in helpfulness and specificity due to a lack of concrete suggestions and feedback. While not perfect, the results align with our goal of generating reviews that are coherent and linkable. See Appendix C for further details on the setup.

In summary, our results suggest that LLM-generated linked document pairs, while sometimes less specific or realistic, offer a practical, scalable way to create sentence-level links that can be used for automatic evaluation, to which we now turn.

5 Automatic Evaluation

5.1 Approaches

With our synthetic data, we test a wide range of retrieval models in a zero-shot setup with the goal of shortlisting top-performing mod-

Model	NEWS-ECB+		NEWS-SYNTH		REVIEWS-SYNTH		REVIEWS-F1000		Overall
	Avg. F1	R@10	Avg. F1	R@10	Avg. F1	R@20	Avg. F1	R@20	Avg. F1
Sparse Models									
BM25	37.26	89.40	32.58	94.35	18.91	66.39	29.53	93.90	29.57
SPLADEv3	39.83	93.23	33.04	95.45	19.46	69.83	29.10	95.59	30.36
BGE-M3-SPARSE	38.81	92.21	34.52	96.49	19.73	70.08	29.68	95.38	30.68
Bi-Encoders									
SFR	43.62	98.01	36.28	97.8	19.20	75.93	25.56	92.05	31.17
all-mpnet-base	42.18	96.52	36.73	98.97	18.09	68.69	24.54	89.16	30.38
BGE-M3-DENSE	42.33	96.23	37.32	99.79	21.52	77.14	29.80	96.14	32.74
Contriever	41.17	95.73	33.17	97.31	17.30	66.69	26.14	93.45	29.45
Dragon+	42.42	96.72	36.99	99.17	21.11	73.50	31.51	97.08	33.01
Cross-Encoders									
BGE-M3-Reranker	41.39	96.74	36.83	100.0	19.52	77.11	27.22	95.74	31.24
ms-marco-MiniLM	41.88	96.27	36.71	98.76	21.26	72.97	31.88	96.65	<u>32.93</u>

Table 2: Performance of retrieval models across synthetic and converted datasets. Metrics are: average F1 score across all cutoffs ($k \in \{1, 3, 5, 7, 10, 20\}$) and recall at a fixed cutoff (R@10 for news, R@20 for reviews). The overall average F1 is computed across all datasets. Recall cutoffs are selected based on domain-specific document lengths to maximize retrieval coverage. Detailed per-dataset and per-cutoff results are provided in Appendix D.

els for the annotation and human evaluation study. The candidate models include sparse models (BM25 (Robertson and Zaragoza, 2009), SPLADEv3 (Lassance et al., 2024), BGE-M3 (Chen et al., 2024a)), bi-encoders (all-mpnet-base-v2⁵, SFR-Embedding-Mistral (Meng et al., 2024), BGE-M3, Contriever (Izacard et al., 2022), Dragon+ (Lin et al., 2023)), and cross-encoders (ms-marco-MiniLM-L6-v2⁶, BGE-M3). Each retriever ranks candidate sentences in the target document based on cosine similarity to a sentence from the source document. We convert ranks into binary decisions via a threshold: all sentences that appear within the top- k are treated as links.

Next, we explore the use of LLMs to refine the retrieval results. We use the best-performing retrieval model to retrieve the top- k candidate target sentences for each source sentence to reduce the search space. We choose the k based on the performance of the retrieval models, and the length of the documents, to maximize the Recall@ k . For NEWS domain datasets, we use top-10 results, and for PEER REVIEW domain, we use top-20. We then use the LLM to classify the source-target pairs as linked or not linked (Figure 2). We further refer to this approach as R+LLM for brevity. We explore two setups:

- **Pairwise:** The LLM is given one source-

target sentence pair at a time from the top- k list to classify them independently.

- **Listwise:** The LLM is given the source sentence and all k target sentences at once, and classifies them in a single pass.

In each setup, we prompt the LLM with both whole documents with four configurations: with a link description, with in-context examples, with both, and with no guidance. For evaluation, we use two open-source models, Phi-4 (14B) (Abdin et al., 2024) and Qwen2.5 (32B) (Yang et al., 2024), selected for their state-of-the-art performance within their respective size classes. We also include GPT-4o⁷, as the SOTA closed-source non-reasoning model at the time of writing, to serve as a high-performance reference point. Prompts templates are given in Appendix E.

5.2 Results

Retriever-only. Table 2 summarizes the performance of all retrieval models across synthetic and converted datasets. The sparse models: BM25, SPLADEv3, and BGE-M3 (Sparse) perform slightly worse overall. Dense models such as SFR, Dragon+, and BGE-M3 (Dense) demonstrate stronger F1 scores across all datasets. Among them, Dragon+ achieves the highest average F1 across datasets and cutoffs, outperforming all other models. This ranking is calculated by averaging the F1 scores computed at each cutoff ($k \in$

⁵<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

⁶<https://huggingface.co/cross-encoder/ms-marco-MiniLM-L6-v2>

⁷Version gpt-4o-2024-08-06

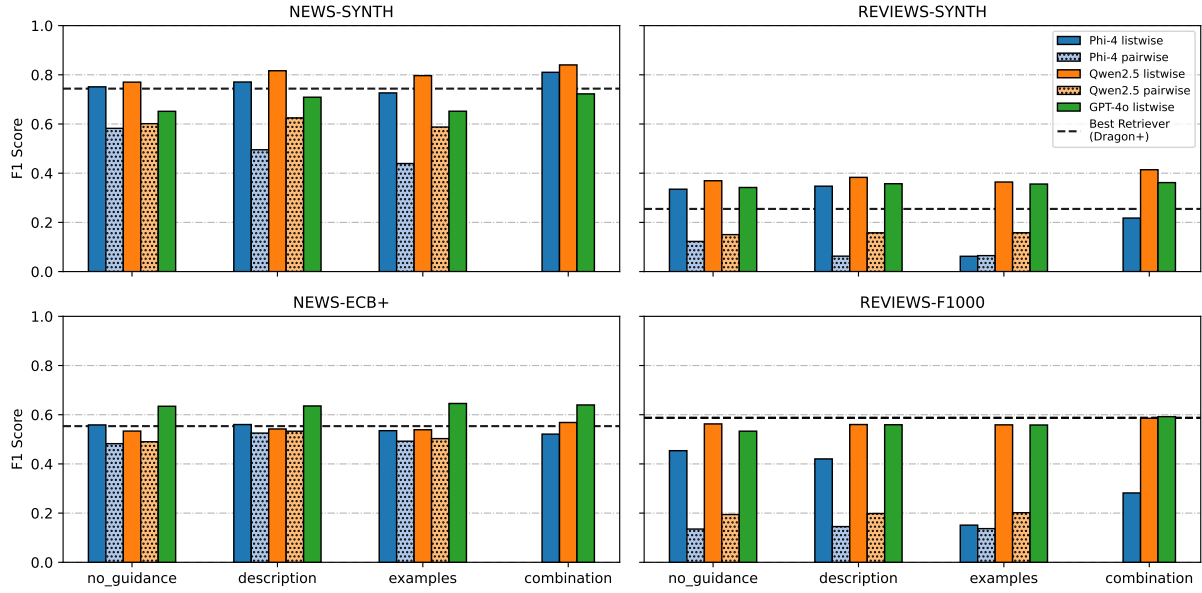


Figure 3: F1 scores across datasets using different prompt configurations and models (Phi-4, Qwen2.5, GPT4o). Results show a consistent trend across domains with listwise prompting outperforming pairwise prompting. Furthermore, combination prompts (tested listwise only) achieve the highest F1 scores on synthetic datasets, while performance gaps narrow on the converted datasets. These results showcase the interaction between prompt design, model capabilities, and dataset complexity.

{1, 3, 5, 7, 10, 20}) for each dataset, followed by averaging across all datasets. While the performance gaps between the top bi-encoders and cross-encoders are not large, Dragon+ is consistently among the strongest, and we select it as the base retriever for subsequent experiments. Notably, while the cross-encoders BGE-M3 (reranker) and ms-marco-MiniLM-L6-v2 also perform competitively, their inference costs are higher compared to Dragon+ which further supports our choice.

R+LLM. Building on the best retriever (Dragon+), we next experiment with using LLMs as classifiers to refine the results. Figure 3 presents the F1 scores of the different R+LLM setups. On the synthetic datasets (NEWS-SYNTH and REVIEWS-SYNTH), applying the LLM as a filter consistently improves performance, particularly for REVIEWS-SYNTH, which benefits from the LLM’s ability to capture nuanced feedback and critique relations. On the converted datasets (NEWS-ECB+ and REVIEWS-F1000), however, the gains are marginal. We attribute this to the inherent mismatch between these datasets and our linking task: NEWS-ECB+ focuses on coreference links, while REVIEWS-F1000 is derived from F1000RD which uses cosine similarity to select candidate links during annotation, naturally favoring simple retrieval-based methods.

Choice of LLM Across the three LLMs tested (Phi-4, Qwen2.5, and GPT-4o), we find that Phi-4 performs on average worse than the others. Qwen2.5 is the strongest overall, especially on the synthetic datasets, while GPT-4o slightly outperforms the others on NEWS-ECB+. We attribute this to properties of the dataset itself: NEWS-ECB+ links are based on coreferent event mentions, which could be easier for stronger models like GPT-4o to detect. Despite these relative differences, all three models achieve only modest absolute F1 scores, in the mid-30s to low-40s on REVIEWS-SYNTH and REVIEWS-F1000, highlighting the intrinsic difficulty of sentence-level linking in long, technical documents. Nevertheless, across all datasets, the best prompting configuration is clear: combining a link description (i.e., a description of what it means for two sentences to be related) with in-context examples leads to the strongest performance. In addition to the prompt content, we find that the way candidate sentences are presented to the LLM matters. In particular, *listwise prompting*, where all top- k candidate sentences are shown together, outperforms *pairwise prompting*, where the LLM considers each candidate in isolation. We hypothesize that listwise prompting allows the LLM to condition its decisions on the global context rather than making them in isolation.

LLM-only ablation. To evaluate the necessity of retrieval, we ablate the retriever and use LLMs alone for classification. We use the best prompt setup (link description + in-context examples; see Appendix F) and instruct the models to classify every sentence in the target document as linked or not, using the full documents as context. Across all datasets, the LLM-only approach underperforms the R+LLM approach, confirming that retrieval is crucial for narrowing the search space and filtering distractors. The performance gap is especially large on the longer more complex peer review documents. For example, GPT-4o reaches only 33.42 and 39.22 F1 on REVIEWS-SYNTH and REVIEWS-F1000, compared to 41.42 and 59.18 with R+LLM. Even on shorter news articles, the LLM-only setup falls short (e.g., 74.27 vs. 84.02 on NEWS-SYNTH). While LLMs can capture some cross-document signals without retrieval, their high inference cost makes them inefficient as standalone solutions compared to their performance, making retrieval as a first step essential for both efficiency and accuracy.

Summary. Taken together, our results suggest that the strongest approach for both domains is R+LLM using Dragon+ and Qwen2.5, prompting listwise with both a link description and in-context examples.

6 Assisted labeling

The final step of our framework applies the best-performing R+LLM approach established on synthetic data to help humans find links in naturally occurring text pairs, and compares it to Dragon+ as a baseline in a human evaluation experiment.

6.1 Setup

For assisted labeling we sample 20 document pairs from each of two application domains: review-paper pairs and news article pairs. **REVIEWS-HE** consists of short conference papers and their corresponding peer reviews, drawn from the ARR22 subset of the NLPeer dataset. We focus on shorter papers to avoid long-context limitations during model inference, and select the shortest available review when multiple are present. **NEWS-HE** comprises article pairs from the SPICED dataset (Shushkevich et al., 2024), sampled from the *Politics*, *Sports*, and *Culture* categories to ensure topical diversity. In both domains, we use spaCy for sentence segmentation, followed by manual corrections to improve segmentation accuracy and remove non-content ele-

ments such as social media links or boilerplate text. See Appendix G for further details on data preparation. To support the annotation process, we used the INCEpTION platform (Klie et al., 2018), which provides a side-by-side document view and allows annotators to accept or reject pre-highlighted candidate links with access to full sentence and document context. See Appendix H for further details on the annotation interface.

6.2 Protocol

Annotators were presented with the source and target documents side-by-side, and asked to evaluate candidate links from the source document to the target using domain-specific guidelines (Appendix K). Candidate links were generated by the best-performing R+LLM approach, the best retriever (Dragon+), and random sampling. In REVIEWS-HE, up to 8 candidate target sentences were shown for each source sentence: top-3 from R+LLM, top-3 from Dragon+, and 2 random distractors. In NEWS-HE, which contains shorter documents, the pool was limited to 5 candidates per source sentence: top-2 from R+LLM, top-2 from Dragon+, and one random distractor. To reduce annotation workload, we excluded source sentences that were too short (three tokens or fewer, typically headers), as well as explicit links (e.g. "Line X" or "Figure Y") which can be resolved trivially (Kuznetsov et al., 2022). If a candidate target sentence is suggested by both R+LLM and Dragon+, it is only displayed once but is counted toward both during the analysis.

6.3 Annotation study

We recruited 15 annotators via Prolific⁸ and conducted a qualification task using one "gold" document pair with 30 cross-document links labeled by the study authors for each domain (Selection criteria detailed in Appendix I). Annotators were provided with annotation guidelines and asked to perform the linking annotation task on these documents. We measured agreement with the gold labels using Cohen’s κ (Cohen, 1960) and selected the two annotators who achieved the highest scores: $\kappa = 0.68$ for REVIEWS-HE and $\kappa = 0.72$ for NEWS-HE, both indicating substantial agreement. The selected annotators then labeled the full evaluation set in two batches of 10 document pairs per domain. After the first batch, we conducted a feedback round to clarify guideline interpretations and

⁸<https://www.prolific.com/>

	REVIEWS-HE	NEWS-HE	Avg.
Annotation Results			
R+LLM only	56.4%	57.0%	56.7%
Retriever only	42.7%	17.8%	30.3%
Both	77.7%	68.6%	73.1%
Random	4.3%	7.7%	6.0%
Statistics			
# Doc. Pairs	20	20	
# Labeled Links	1022	804	

Table 3: Agreement rate between the two annotators on candidate links, broken down by link suggestion method. Also statistics on the number of document pairs and total labeled links for each domain.

highlight common issues, after which the annotation resumed independently. Final inter-annotator agreement across both batches was $\kappa = 0.59$ for REVIEWS-HE and $\kappa = 0.60$ for NEWS-HE, reflecting substantial agreement given the subjective and open-ended nature of the task. On reviews, this exceeds the reported agreement in machine-assisted annotation by Kuznetsov et al. (2022), despite our use of crowd annotators with minimal training.

6.4 Results and Analysis

Table 3 summarizes the acceptance rates across link suggestion methods. These results provide validation for our framework and align closely with trends observed on synthetic data. As shown in Table 3, links suggested by both Dragon+ and the R+LLM achieve the highest acceptance rates: 77.7% in REVIEWS-HE and 68.6% in NEWS-HE. This is expected, as mutual agreement between both indicates higher confidence in the selection.

More importantly, we observe a consistent pattern across domains: links selected only by the R+LLM approach are accepted at substantially higher rates than those selected only by Dragon+, namely 56.7% vs. 30.3% on average. The contrast is especially pronounced in the news domain (57.0% vs. 17.8%), where retrieval alone appears less effective. We attribute this to the retriever’s reliance on semantic similarity which fails to capture the nuanced, context-sensitive understanding required for accurate linking. In contrast, LLMs can reason about discourse and topic structure, drawing on prior knowledge to identify implicit connections. This highlights the LLM’s strength as a task-aware filter to refine surface-level retrieval results.

Crucially, this finding echoes our results on synthetic data, where adding LLM classification con-

sistently improved performance. The human evaluation confirms that these gains hold on natural text, demonstrating that the LLM contributes meaningful additional value beyond the retriever’s capabilities. The low acceptance rate for random distractor candidates (6.0% on average) confirms that the task is non-trivial and that valid cross-document links are unlikely to arise by chance. This underscores the importance of LLM-guided candidate selection.

Finally, our study results in substantially-sized human-labeled annotated datasets in peer reviews, and news articles (Table 3). Each document pair is associated with one set of links per annotator, and the corpora are comparable in size to other datasets form the literature. For example, in the reviews domain, our data features 10 times more annotated links than (D’Arcy et al., 2024), and is comparable in size to the expert annotated subset of F1000RD. This data enables evaluation of future systems for review-paper linking and news linking, and allows the study of NLP applications for peer reviewing assistance, media framing, and beyond. We leave this exploration for future work.

7 Conclusion

High annotation cost and the diversity of cross-document links are a key barrier toward wide application of cross-document linking. We proposed a general, task- and domain-agnostic framework for sentence-level cross-document linking that reduces the cost and complexity of annotation. By using LLMs to generate synthetic linked document pairs, we enabled automatic evaluation for a wide range of linking approaches. The best-performing approaches were validated via human evaluation to identify the best approach without requiring labeled data upfront, and producing labeled data as a secondary outcome.

Our experiments in peer review and news domains have shown that combining retrieval with LLM-based classification yields highest-quality link suggestions. The framework generalizes across link types and domains, allowing scalable dataset construction and evaluation of linking approaches. By reducing the cost and complexity of annotation, we lower the barrier for studying a wide range of link types and domains. Our datasets of linked natural document pairs pave the path to applied studies of cross-document understanding in peer reviewing assistance, news and media framing analysis, and other application domains.

Ethical considerations

We believe that our work does not carry ethical risks and contributes to better technological support for many socially relevant application areas such as academic quality control, journalism, fake news detection and propaganda analysis. The participants of the synthetic data validation study in the news domain contributed on a voluntary basis; the participants of the synthetic data validation study on peer reviews conducted it as a part of their employment at the authors’ institution. The crowdworkers on Prolific have been fairly compensated with 13 Euro per hour. The NLPeer dataset has a CC BY 4.0 license, and we release our corresponding derivative dataset under the same license. The ECB+ corpus has a CC BY 3.0 license which allows us to release the corresponding derivative dataset under CC BY 4.0 license. The SPICED dataset has a CC BY 4.0 license, but the licensing for its source material (news) is unclear. To work around this, we release a script for reconstructing the dataset along with our added data. All datasets were used according to their intended research purpose.

Limitations

Synthetic data generation. Our synthetic data relies on prompt-based generation conditioned on the target document. While our results suggest that this approach produces high quality data, future work could improve the coverage by prompting for individual sentence-level links or using multi-pass generation strategies.

Model coverage. Due to the cost of LLM experiments, we needed to trade off between the number of evaluations per LLM and the number of LLMs considered. We chose breadth, focusing on a representative set of retrievers and LLMs. Expanding to more models, prompt variants, and multiple runs would offer deeper insight into robustness and prompt sensitivity, and is left for future work.

Document modalities. Our framework currently operates on text only. In domains like scientific writing, non-textual elements (e.g., tables, figures) often contain crucial information that can be linked to and linked from. Multi-modal linking is an exciting direction with its own line of related work, and we leave the application of our framework to this case to the follow-up studies.

Link coverage. While our framework allows measuring the precision of link suggestions, it does not guarantee true recall measurement. This principal limitation is common in prior work on linking-related tasks: Kuznetsov et al. (2022) limit the number of candidates per link to 5, while D’Arcy et al. (2024) only focus on linking review comments to *edits* in the paper in response to them. We mitigate this issue by retrieving a large candidate pool before classification, and by evaluating candidate approaches on fully linked synthetic data, but ultimately we prioritize annotation efficiency over exhaustiveness. Measuring true linking recall in absence of exhaustively labeled annotated gold data remains an open challenge to be tackled in the future.

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A Datasets Conversion

In **NEWS-ECB**, for each document pair, we label a sentence pair as linked if they contain event mentions in the same cluster. While this conversion provides a useful signal, it introduces two key limitations: (1) the resulting labels are incomplete, as only coreferent events are annotated, and (2) some event mentions occur in headlines or titles, which leads to spurious links from titles to many other sentences. In **REVIEWS-F1000**, the authors didn’t adjudicate the annotations, so we only select examples where both expert annotators agreed on the label. This reduces the number of links we can use, but ensures we have a comparable dataset to **REVIEWS-SYNTH**.

B Quantitative Criteria

To assess how closely our synthetic documents resemble natural ones, we selected three document-level metrics that capture distinct stylistic dimensions: (1) lexical diversity, as a proxy for richness of vocabulary; (2) subjectivity, to assess differences in tone or stance; and (3) lexical complexity, using the Flesch-Kincaid Reading Ease score, to estimate linguistic difficulty. These criteria were chosen to reflect qualities that may affect linkability and document coherence, while remaining agnostic to content. Figure 13 plots the results for synthetic and natural documents in both domains.

C Rationale for Evaluation Criteria

We selected domain-specific evaluation criteria to reflect both general text quality and the particular requirements of our linking task. For **NEWS-SYNTH**, we assessed:

- **Fluency** assesses grammatical correctness and naturalness of the language, ensuring the text is readable.
- **Coherence** measures logical flow and topic consistency, which is critical for linking tasks that depend on understanding article structure and progression.
- **Realism** evaluates whether the article could plausibly have been written by a human journalist, and ensure the generated article is not clearly AI-generated.
- **Specificity** checks for the inclusion of concrete facts (e.g., names, dates, events, etc.) to ensure the AI-generated review is not general and vague.

For **REVIEWS-SYNTH**, we assessed:

- **Fluency and coherence**, as above, ensure the review reads naturally and follows a logical structure.
- **Helpfulness** replaces realism in this context, measuring whether the review provides meaningful feedback that could assist authors, which is central to the function of peer reviews.
- **Specificity** captures whether the review addresses concrete elements of the paper, such as specific sections, claims, or experiments, helping differentiate generic summaries from insightful critique.

These criteria were chosen to ensure that our synthetic datasets maintain linguistic quality and task relevance, even if they do not replicate all the nuanced characteristics of human writing. Figure 14 illustrates detailed validation results.

D Retrievers’ Results

Tables 6, 7, 8 and 9 provide the full retrieval results for all models for all datasets individually, broken down by cutoff values ($k \in \{1, 3, 5, 7, 10, 20\}$).

E LLM Prompting Techniques

We created two prompt templates corresponding to the *pairwise* and *listwise* classification setups (Figures 4 and 5). Each prompt includes a system message that frames the task: determining whether a candidate target sentence from one document is related to a source sentence from another document, using full document context. In the pairwise setup, the model classifies one sentence pair at a time; in the listwise setup, it classifies a ranked list of candidate targets in a single pass. Both formats support four prompting configurations: no guidance, link description only, in-context examples only, and both. The prompts are designed to output binary classifications and return structured JSON outputs, facilitating scalable evaluation across top- k retrieved candidates. For Phi-4 and Qwen2.5, we used vLLM (Kwon et al., 2023) with the structured outputs function to ensure output consistency. The experiments with the open-source models were run on A100 GPUs. For GPT-4o, we used the OpenAI API with structured outputs too. For all models, we set $\text{temperature} = 0.3$ and $\text{top-p} = 0.9$.

F LLM-only Ablation

Figure 7 shows the prompt template used in the LLM-only setup, where the model classifies all

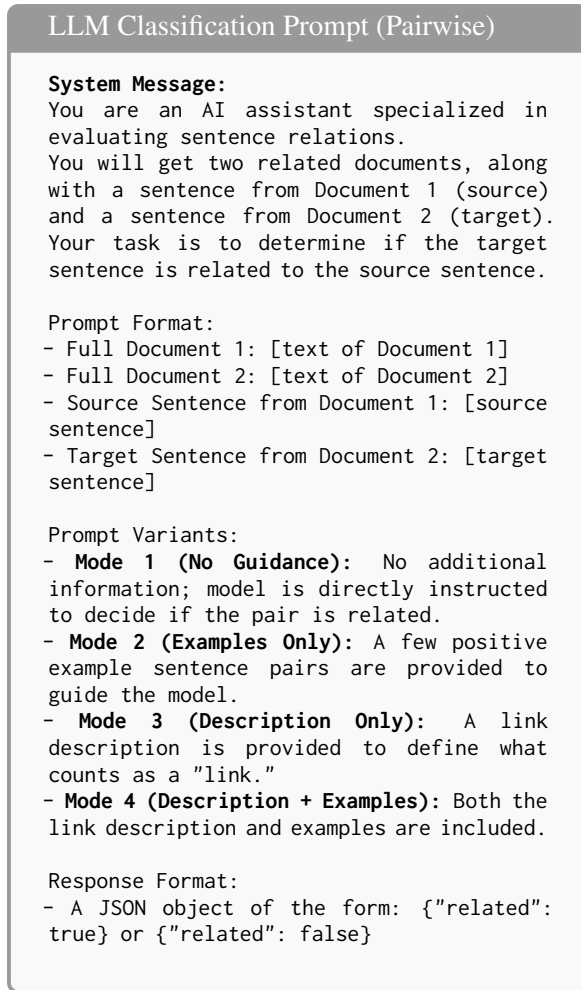


Figure 4: Prompt template for LLM-based pairwise sentence classification.

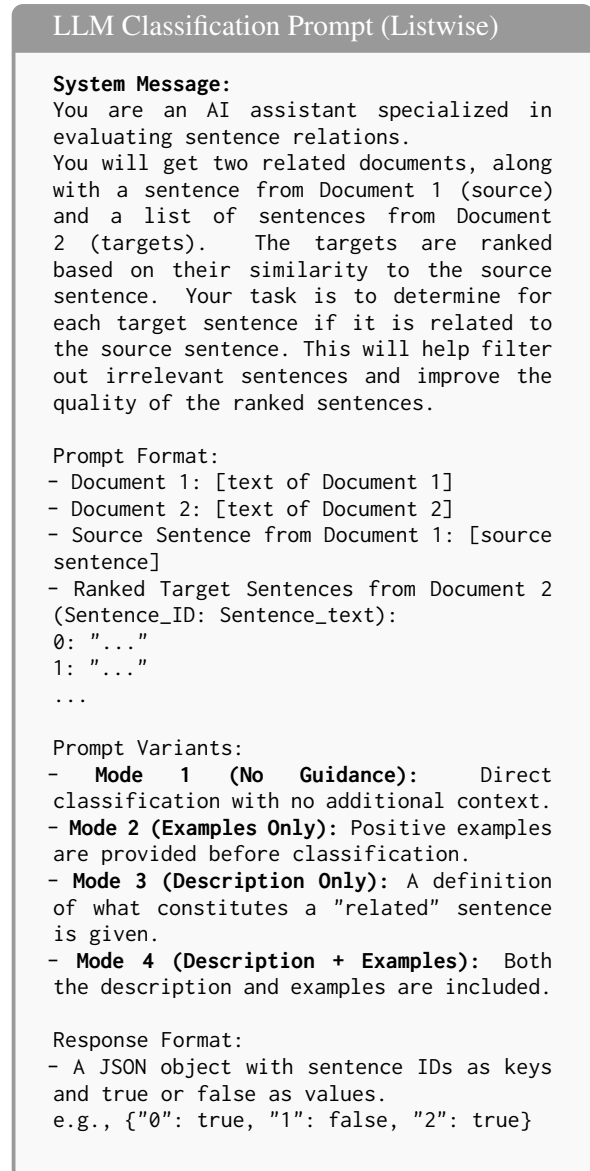


Figure 5: Prompt template for LLM-based listwise sentence classification.

sentences in the target document without prior retrieval. The prompt includes full document context and a single source sentence, and instructs the LLM to evaluate each target sentence for its relevance. As in the other setups, we support four prompt configurations: no guidance, with in-context examples, with a link description, and with both. We used structured outputs here too, with temperature = 0.3 and top-p = 0.9

G Data Preparation for Human Evaluation

For REVIEWS-HE, we randomly sampled 20 paper-review pairs from the ARR22 subset of NLPeer from the shortest 100 ones to minimize long-context issues during model inference. In cases with multiple reviews, we selected the shortest available. Peer reviews were segmented into sentences using spaCy and manually corrected for sen-

tence boundary errors. The paper texts came pre-segmented via NLPeer. For NEWS-HE, we sampled 20 article pairs from the SPICED dataset across the *Politics*, *Sports*, and *Culture* categories to ensure diversity. The SPICED dataset is a multi-document summarization dataset that pairs news articles of different sources that talk about the same topic or event, so the document pairs are naturally related, and suitable for our task.

H Annotation interface

We used the INCEpTION annotation platform (Klie et al., 2018), which supports side-by-side viewing of two documents and interactive annota-

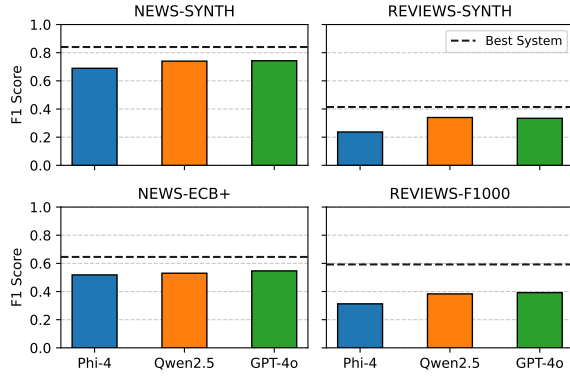


Figure 6: LLM-only ablation. Models were prompted with both full documents (source and target), the specific source sentence, a link description, and in-context examples. While this setup captures more context and task-specific information, it still underperforms compared to the combination of Retriever and LLM. This highlights the importance of retrieval for narrowing the candidate space and reducing distractors, especially in long documents.

tion (Figure 15). Source sentences, along with their candidate target sentences were pre-highlighted in the interface, along with visual cues, in the form of arrows pointing from the source sentence to each target sentence, to reinforce the linking aspect of the task. Annotators could view the full sentence and surrounding context and were asked to accept or reject each candidate. Candidates from both the retriever and R+LLM approach were visually indistinguishable and shown in the normal order they show up in the document.

I Annotator Selection Criteria

Annotators were screened based on the following eligibility criteria: Native English speakers, residing in United States, United Kingdom, Canada, Australia, or Ireland, with at least a Master’s or PhD degree in Computer Science, a Prolific approval rate of at least 90%, and had completed a minimum of 30 prior studies on the platform.

J Agreement trends and Annotation Feedback.

We observed a learning curve in annotator agreement. In REVIEWS-HE, agreement started relatively low, improved steadily toward the middle of the batch, and declined slightly toward the end, likely due to annotator fatigue. In NEWS-HE, agreement rose more gradually and plateaued at a slightly lower level overall. First-batch inter-annotator

LLM-only Classification Prompt

System Message:

You are an AI assistant specialized in evaluating sentence relations. You will get two related documents, along with a sentence from Document 1 (source). Your task is to determine for each sentence in the target document (Document 2) if it is related to the source sentence. This will help filter out irrelevant sentences.

Prompt Format:

- Document 1: [text of Document 1]
- Document 2: [text of Document 2]
- Source Sentence from Document 1: [source sentence]

Prompt Variants:

- **Mode 1 (Minimal):** No additional context—just classify based on the input documents and source sentence.
- **Mode 2 (Examples Only):** A few examples of positive sentence pairs are provided.
- **Mode 3 (Description Only):** A link description is provided to define what counts as a link.
- **Mode 4 (Description + Examples):** Both the link description and in-context examples are included.

Response Format:

- A JSON object where each key is a target sentence ID and each value is either true or false.
e.g., {"11": true, "4": false, "9": true}
- Ensure that every sentence in Document 2 is classified.

Figure 7: Prompt template for LLM-only sentence classification across an entire target document.

agreement was $\kappa = 0.55$ for REVIEWS-HE and $\kappa = 0.54$ for NEWS-HE, indicating moderate agreement despite the subjective nature of the task.

After the first batch, we reviewed disagreement patterns. One annotator tended to reject links that relied on broader document context, favoring self-contained links. The other applied stricter interpretations of the guidelines, occasionally rejecting links that only partially aligned. Based on these observations, we gave targeted feedback, clarified borderline cases, and encouraged consistent application of the criteria. This led to improved agreement in the second batch ($\kappa = 0.62$ for REVIEWS-HE, 0.64 for NEWS-HE). However, this could also be attributed to the annotators getting used to the task, and thus understanding the guidelines better.

Link	No Link
The review sentence critiques or praises the same claim or result as the paper sentence.	The review mentions grammar, structure, or other writing aspects not discussed in the candidate sentence.
The review analyzes or extends a method/result introduced in the paper sentence.	The review proposes future work or experiments not mentioned in the paper sentence.
The review addresses the same technical method, component, or result.	The review discusses different topics or experiments from the candidate sentence.

Table 4: Linking Criteria for Peer Reviews

Link	No Link
Both sentences refer to the same event or topic.	Sentences refer to different events, even if they happen at the same place/time.
One sentence is a follow-up or elaboration of the other.	The target sentence provides unrelated opinion, editorial, or background.
One is a paraphrase or restatement of the other (even if phrased differently).	The anchor and target describe distinct stories with no direct overlap.

Table 5: Linking Criteria for News

K Annotation Guidelines

Table 4 and 5 contain the criteria provided to the annotators to decide whether a sentence pair should be labeled as linked or not. Because the concept of a "link" can be broad, we provided domain-specific examples to clarify what qualifies as a meaningful connection in the context of peer reviews and news articles.

L Synthetic Data Generation Prompts

To generate synthetic peer reviews and news articles, we designed detailed prompting templates that reflect realistic domain practices while introducing controlled variation. The peer review prompt (Figure 9) instructs the model to simulate a structured review grounded partially in the source text, while maintaining a natural and critical tone consistent with academic peer reviews. It emphasizes abstraction, editorial judgment, and partial grounding to avoid simple paraphrasing. Because the news articles from the WikinewsSum dataset were scrapped directly from the webpage, they contain some artefacts that are not content of the article. Thus, we first clean that using GPT-4o-mini with

Synthetic Peer Review Generation Prompt

This is a scrapped article from Wikinews. Due to the scrapping, it may contain sentences that are image captions and social media links. Remove such sentences, but do not change the content of the article. Output the cleaned article in JSON format, where the key is 'cleaned_article' and the value is the cleaned article text.
Input: {INPUT}

Figure 8: Prompt template for corrected scrapped news articles.

the prompt template shown in Figure 8. Next, the generation prompt (Figure 10) guides the model to produce synthetic news article covering the same topic, but differing in structure, tone, and emphasis to simulate diverse editorial styles. Both generation prompts include format specifications and grounding strategies. While these prompts may not be the most optimal, they produce sufficiently high-quality and controllable outputs for our specific goal of in the linking task. Table 10 provides illustrative examples of generated sentences and their corresponding linked natural sentences in both domains

M Human Evaluation Results

Figures 11 and 12 illustrate the results of the human evaluation in detail.

Synthetic Peer Review Generation Prompt

Task Overview

You are a peer reviewer evaluating a research paper in NLP. Your task is to write a realistic and well-rounded peer review for the paper.

Guidelines:

- Your review should be structured and natural, consisting of 8-12 sentences.
- 3 to 5 sentences should be implicitly grounded in specific ideas from the paper. These should express relevant critiques, observations, or praises without directly quoting or referencing the original text.
- The remaining sentences should address broader aspects such as clarity, methodology, contributions, generalization, writing quality, or suggestions for improvement.
- Do NOT explicitly cite, reference, or quote any sentence from the paper. The review should not be a direct commentary on specific lines.

How to Structure the Review:

- Rephrase, summarize, or abstract ideas: When addressing parts of the paper, reword and generalize instead of copying.
- Introduce new considerations: Some comments should reflect editorial judgment, unanswered questions, or high-level concerns rather than being tied to specific sentences.
- Omit some possible links: Not every review sentence should directly correspond to a sentence from the paper. Aim for a balanced mix of specific and general feedback.
- Rearrange information: The review's flow should be different from the order of the paper to reflect a natural peer review process.

Input Format

You will receive a JSON object where:

- Each key is a sentence index from a paper section(s).
- Each value is the corresponding sentence text.

Output Format

- A peer review as a JSON object where:
 - Each key is a sentence index in the review (starting from 0).
 - Each value is the corresponding sentence text.
- A sentence mapping that links review sentences to the paper as a JSON object where:
 - Keys: Indices from the peer review.
 - Values: A list of corresponding indices from the paper section(s), or null if the sentence is not directly linked.

Example Input:{INPUT_EXAMPLE}

Example Output:{OUTPUT_EXAMPLE}

Input:{INPUT}

Output:

Figure 9: Prompt template for generating REVIEWS-SYNTH.

Model	K=1			K=3			K=5			K=7			K=10			K=20		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
bm25	56.71	34.12	40.91	37.21	63.17	45.03	28.75	76.74	40.28	24.17	83.59	35.95	21.17	89.40	32.43	18.72	96.40	28.95
splade	62.03	37.47	44.87	41.60	70.11	50.25	30.74	81.93	43.09	25.44	88.37	37.92	21.88	93.23	33.60	18.90	98.26	29.27
bgem3-sparse	59.20	35.30	42.47	39.94	67.59	48.30	30.07	80.30	42.17	25.07	86.97	37.35	21.70	92.21	33.31	18.88	98.07	29.24
sfr	69.14	41.03	49.50	48.34	80.72	58.19	34.51	91.97	48.38	27.49	95.85	41.04	22.81	98.01	35.13	19.02	99.68	29.50
all-mpnet	65.08	39.40	47.12	45.81	77.36	55.35	33.23	88.90	46.62	26.76	93.52	39.95	22.48	96.52	34.60	18.99	99.36	29.44
bgem3-dense	67.65	41.05	49.06	45.55	76.86	55.01	33.03	88.13	46.32	26.59	92.78	39.69	22.44	96.23	34.53	18.95	98.95	29.38
contriever	64.13	38.85	46.46	43.38	73.45	52.46	32.16	85.82	45.10	26.26	91.65	39.18	22.35	95.73	34.38	18.99	99.34	29.45
dragon_plus	66.67	40.31	48.21	45.82	77.36	55.36	33.27	89.02	46.68	26.86	93.87	40.10	22.53	96.72	34.67	19.01	99.50	29.48
bge-reranker	61.29	36.29	43.78	44.86	75.38	54.10	33.03	88.05	46.29	26.78	93.33	39.96	22.56	96.74	34.72	19.02	99.69	29.50
ms_marco_minilm	65.06	39.03	46.83	45.24	76.24	54.63	32.84	87.76	46.07	26.66	93.07	39.80	22.45	96.27	34.55	18.98	99.25	29.43

Table 6: Results on NEWS-ECB+. Precision (P), Recall (R), and F1 at different cutoffs (K).

Synthetic News Generation Prompt

Task Overview:

You are generating a news article that covers the same event or topic as an original article presenting it from a different editorial angle. The goal is to simulate how separate news organizations might independently report on the same subject, differing in structure, tone, detail, and emphasis. The new article should be a plausible alternative version of coverage on the same topic, not a direct rephrasing or summary of the original.

Guidelines:

- The article should be realistic, coherent, and reflective of a distinctive voice or editorial style.
- 3 to 5 sentences in the new article should reflect content from the original. These sentences may describe similar facts, events, or issues, but using different wording, tone, or framing.
- Do not replicate the original article's sentence-by-sentence structure or closely paraphrase its content.
- The remaining sentences should introduce: New but plausible perspectives, context, or editorial framing. Additional background or expert input. A different narrative structure or omission of certain original points.
- The new article must have a different length from the original, but not be much shorter than the original.

Variation Strategies:

- Rephrase and shift style: Change vocabulary, sentence structure, or writing tone to reflect a different editorial voice.
- Frame the topic differently: Adjust emphasis or viewpoint-, for example, highlighting controversy, local impact, or long-term implications.
- Add or omit information: Introduce plausible context, background, or expert input, or skip less relevant details from the original.
- Reorganize the narrative: Present the information in a different order to create a new logical or rhetorical flow.

Input Format:

You will receive a JSON object where:

- Each key is a sentence index from the original article.
- Each value is the corresponding sentence text.

Output Format:

- The generated news article as a JSON object:
- Keys: Indices of sentences in the new article (starting from 0).
- Values: The text of each sentence.

A sentence mapping that links sentences in the new article to sentences in the original article as a JSON object where:

- Keys: Indices of sentences in the new article.
- Values: A list of sentence indices from the original that the new sentence relates to, or null if it is not directly linked to any original sentence.

Example input: {INPUT_EXAMPLE}

Example Output: {OUTPUT_EXAMPLE}

Input: {INPUT}

Output:

Figure 10: Prompt template for generating NEWS-SYNTH.

Model	K=1			K=3			K=5			K=7			K=10			K=20		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
bm25	65.29	60.26	61.78	29.75	78.60	42.25	19.75	86.12	31.52	14.64	89.23	24.73	11.14	94.35	19.63	8.63	99.59	15.56
splade	67.36	61.57	63.36	29.48	78.39	42.00	20.17	87.16	32.08	15.05	91.43	25.41	11.26	95.45	19.85	8.61	99.17	15.52
bgem3-sparse	71.49	65.85	67.56	31.96	84.86	45.50	20.41	88.95	32.56	15.35	93.03	25.89	11.39	96.49	20.07	8.63	99.59	15.56
sfr	74.79	67.99	70.04	34.71	90.11	48.91	22.31	95.23	35.35	16.29	97.18	27.37	11.59	97.80	20.42	8.65	100.00	15.60
all-mpnet	76.86	70.47	72.38	34.57	90.79	48.99	22.15	96.17	35.30	16.35	97.85	27.50	11.72	98.97	20.64	8.65	100.00	15.60
bgem3-dense	79.34	72.53	74.59	35.26	92.11	49.83	22.23	95.43	35.30	16.53	99.04	27.80	11.80	99.79	20.79	8.65	100.00	15.60
contriever	65.70	60.48	62.05	30.58	81.16	43.51	19.92	86.68	31.76	15.35	92.27	25.84	11.51	97.31	20.27	8.65	100.00	15.60
dragon_plus	78.51	72.52	74.38	34.71	91.54	49.29	21.57	93.90	34.42	16.41	98.47	27.61	11.72	99.17	20.64	8.65	100.00	15.60
bge-reranker	76.03	70.12	71.90	34.85	90.66	49.17	22.40	95.85	35.51	16.65	99.17	27.96	11.84	100.00	20.86	8.65	100.00	15.60
ms_marco_minilm	82.64	75.56	77.69	33.88	88.79	47.98	21.90	94.46	34.83	16.12	96.75	27.13	11.72	98.76	20.63	8.65	100.00	15.60

Table 7: Results on NEWS-SYNTH. Precision (P), Recall (R), and F1 at different cutoffs (K).

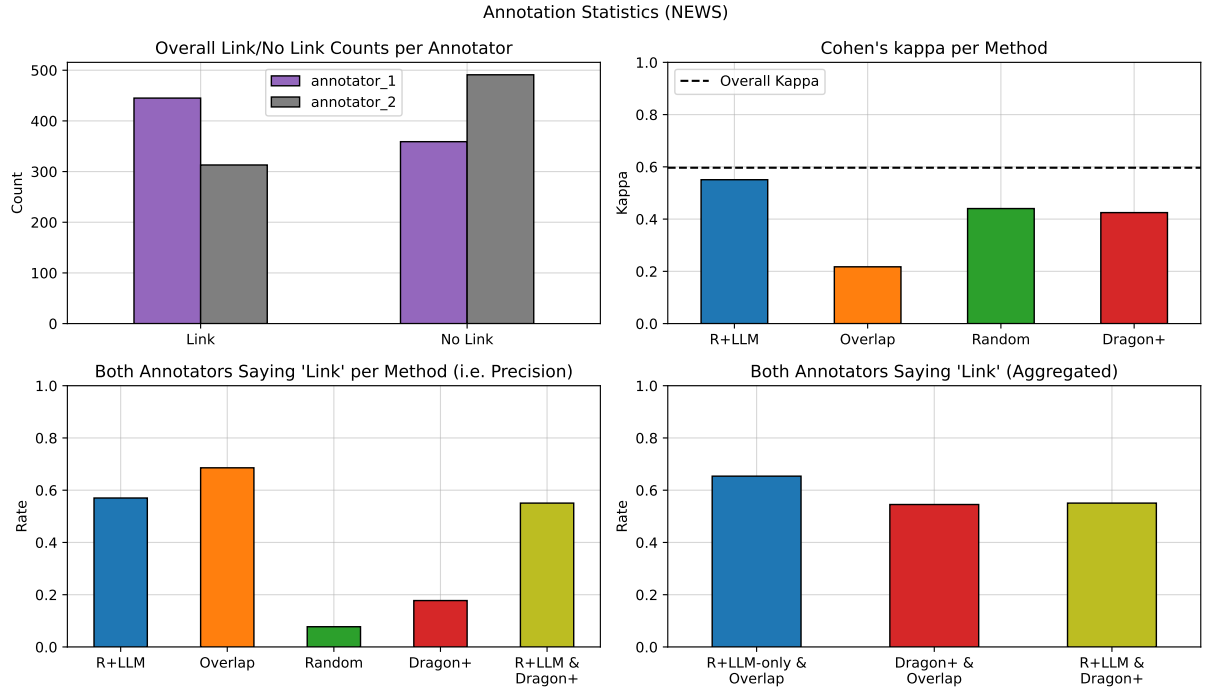


Figure 11: Annotation statistics in NEWS-HE.

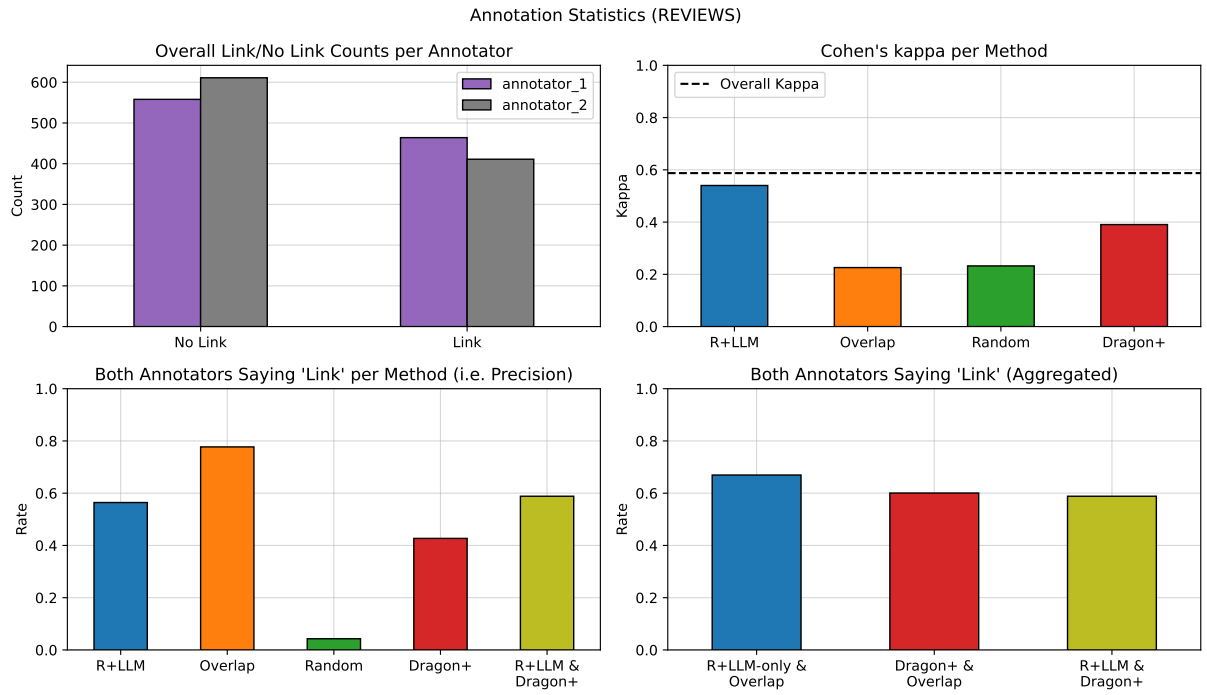
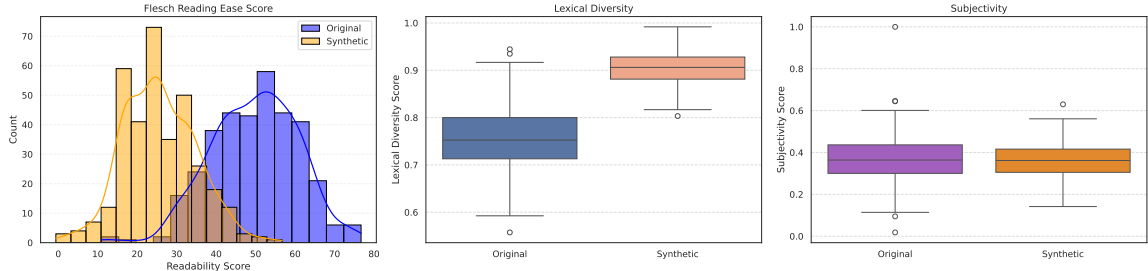
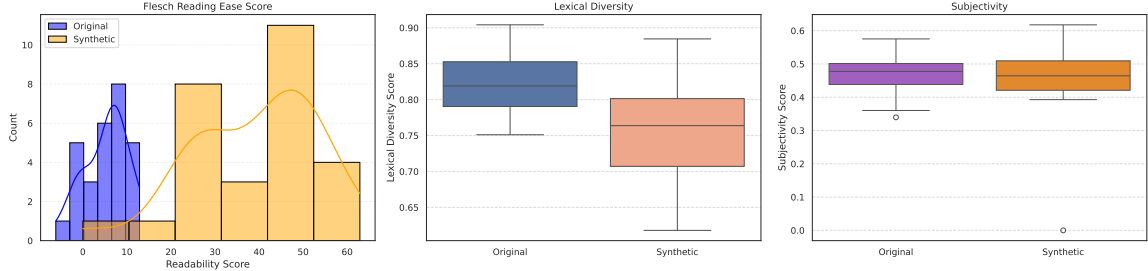


Figure 12: Annotation statistics in REVIEWS-HE.

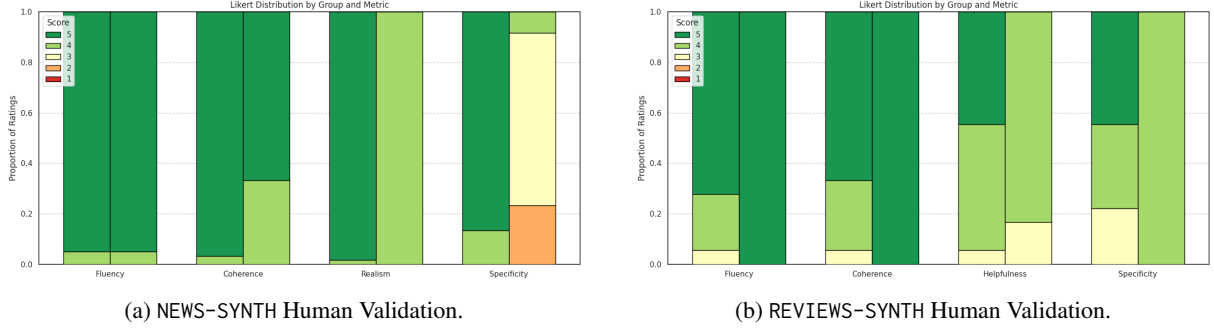


(a) Quantitative comparison of synthetic news to natural news data.



(b) Quantitative comparison of synthetic reviews to natural reviews.

Figure 13: Comparison of synthetic and natural documents in terms of stylistic and structural metrics, across two domains.



(a) NEWS-SYNTH Human Validation.

(b) REVIEWS-SYNTH Human Validation.

Figure 14: Human evaluation results for NEWS-SYNTH (left) and REVIEWS-SYNTH (right).

Model	K=1			K=3			K=5			K=7			K=10			K=20		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
bm25	28.63	16.70	19.93	19.24	30.57	22.21	16.12	41.64	22.03	13.09	47.14	19.57	10.97	55.01	17.64	6.83	66.39	12.11
splade	25.11	14.49	17.27	20.85	33.58	24.16	16.56	42.65	22.66	14.10	51.12	21.12	11.72	60.04	18.92	7.14	69.83	12.65
bgem3-sparse	26.87	14.79	17.93	22.17	36.03	25.79	16.74	44.47	23.09	13.91	49.90	20.79	11.32	56.08	18.15	7.11	70.08	12.62
sfr	24.67	13.19	16.20	20.12	31.34	23.09	16.56	43.01	22.64	13.97	49.70	20.81	11.59	58.52	18.64	7.80	75.93	13.82
all-mpnet	27.75	15.10	18.40	17.77	27.94	20.31	15.07	38.93	20.58	12.96	46.37	19.36	10.88	54.90	17.52	6.96	68.69	12.36
bgem3-dense	29.96	16.56	20.15	22.32	35.17	25.73	18.68	47.51	25.48	15.61	55.51	23.31	12.78	64.00	20.53	7.84	77.14	13.91
contriever	24.67	13.27	16.34	18.21	28.16	20.88	14.10	36.51	19.33	12.59	45.04	18.86	10.18	51.32	16.43	6.74	66.69	11.98
dragon-plus	30.84	17.78	21.28	21.73	36.03	25.48	18.33	48.69	25.24	15.10	55.14	22.65	11.63	59.28	18.74	7.49	73.50	13.29
bge-reranker	23.35	12.34	15.18	20.41	34.16	23.98	16.12	43.83	22.33	14.73	53.11	22.06	12.16	60.70	19.54	7.91	77.11	14.02
ms_marco_minilm	33.92	20.31	23.96	23.64	38.66	27.60	18.59	48.73	25.52	15.10	53.58	22.53	11.94	59.51	19.19	7.42	72.97	13.17

Table 8: Results on REVIEWS-SYNTH. Precision (P), Recall (R), and F1 at different cutoffs (K).

Model	K=1			K=3			K=5			K=7			K=10			K=20		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
bm25	58.28	53.15	54.75	28.21	73.84	40.00	18.75	81.30	29.95	14.15	85.36	23.91	10.29	88.51	18.22	5.51	93.90	10.33
splade	54.36	49.33	50.91	28.31	73.68	40.06	18.94	81.04	30.15	14.30	85.51	24.12	10.66	90.47	18.82	5.64	95.59	10.57
bgem3-sparse	57.49	52.53	54.11	28.57	74.61	40.49	18.94	81.68	30.23	14.31	85.99	24.17	10.49	89.20	18.53	5.62	95.38	10.54
sfr	45.94	40.83	42.44	24.39	62.65	34.34	16.96	72.01	26.94	13.08	77.95	22.05	9.82	82.82	17.34	5.46	92.05	10.23
all-mpnet	43.98	39.56	40.94	23.21	60.00	32.76	16.16	69.13	25.71	12.59	75.13	21.23	9.46	80.34	16.71	5.26	89.16	9.86
bgem3-dense	57.30	51.52	53.33	28.80	74.30	40.62	19.28	82.02	30.62	14.64	87.07	24.66	10.72	91.02	18.94	5.67	96.14	10.63
contriever	46.91	42.36	43.79	24.49	63.80	34.68	17.43	74.47	27.73	13.40	79.65	22.58	10.06	85.49	17.77	5.50	93.45	10.31
dragon_plus	63.08	56.75	58.73	30.33	77.79	42.67	20.20	86.17	32.11	15.10	89.80	25.44	10.95	93.02	19.35	5.73	97.08	10.74
bge-reranker	47.50	43.42	44.71	26.15	68.26	37.06	18.20	78.30	29.02	13.92	83.32	23.50	10.43	88.60	18.43	5.65	95.74	10.58
ms_marco_minilm	64.94	59.24	61.03	30.43	79.37	43.09	20.12	86.37	32.04	15.00	89.68	25.30	10.84	92.41	19.16	5.68	96.65	10.65

Table 9: Results on REVIEWS-F1000. Precision (P), Recall (R), and F1 at different cutoffs (K).

Domain	Source Sentence (Generated)	Target Sentence (Natural)
NEWS	In Southern Europe, the ripple effects of the disrupted flight schedules are being felt widely, with airlines cautioning that recovery from this crisis could take days or even weeks.	Some flights from Spain and Portugal, together with upwards of 4,000 flights across Northern Europe, have been affected, and the knock-on effect of aircraft and crews out of position could disrupt air travel worldwide for up to 72 hours.
NEWS	Newsweek’s Steven Levy has pointed out that top-ranked blogs remain overwhelmingly white and male, underscoring a digital divide as real as any glass ceiling.	Steven Levy, a senior editor of Newsweek, has recently written a column addressing concerns about the over-representation of white males among top bloggers on the Internet.
PEER-REVIEWS	The inclusion of negative samples in the supervised fine-tuning pipeline appears to be non-trivial, as this variant underperforms relative to other supervised strategies.	Among all supervised methods, the SFT with negatives performs the worst, showing that using negative feedback in supervised training analogically to preference optimization is non-trivial.
PEER-REVIEWS	Coverage limitations of some rule-based components mean that a sizable fraction of utterances remain unhandled, which may impact the overall robustness.	P1.1 and P2 have to meet certain conditions to be applied, therefore they do not have full coverage of CS data: 36% and 31% for SEAME, 60% and 58% for Miami.

Table 10: Examples of sentence-level links in the NEWS-SYNTH and REVIEWS-SYNTH datasets. Source sentences are generated using an LLM conditioned on the target document, with sentence-level links specified during generation.

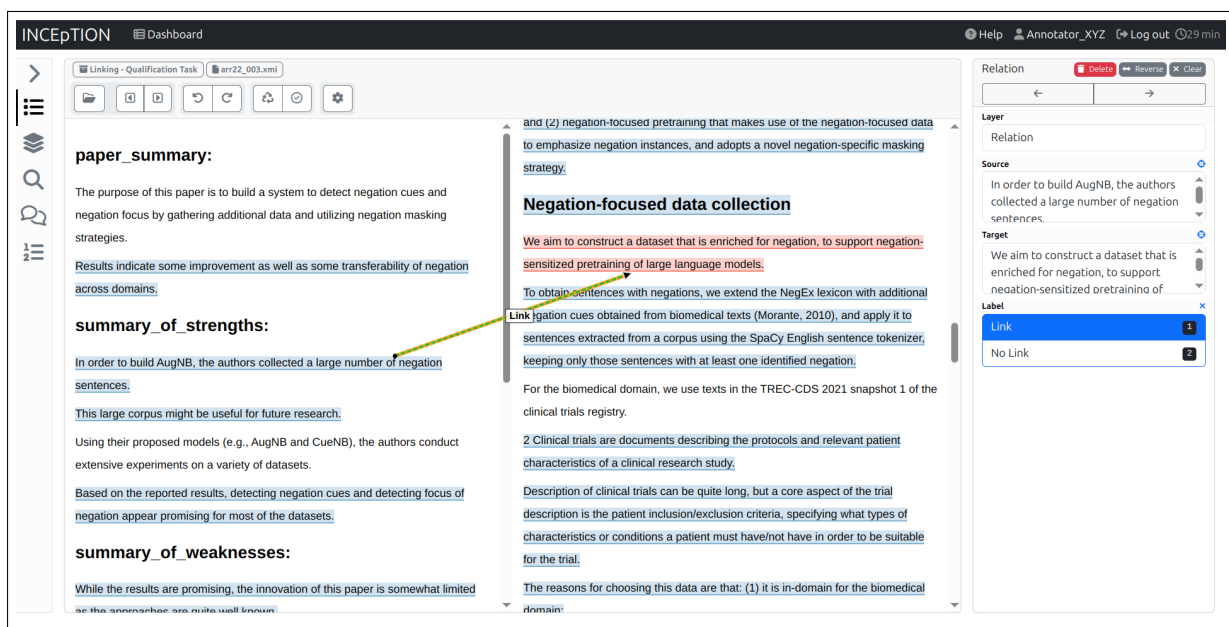


Figure 15: Annotation Interface.